Vida Jamali

Assistant Professor | Dr. James Robert and Margaret Spencer Early Career Fellow
School of Chemical and Biomolecular Engineering, Georgia Institute of Technology
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PROFESSIONAL APPOINTMENTS

Aug. 2022–present	**Georgia Institute of Technology**, Atlanta, GA Assistant Professor, School of Chemical and Biomolecular Engineering
Apr. 2026–present	Dr. James Robert and Margaret Spencer Early Career Fellow Georgia Institute of Technology
Aug. 2022–present	Daniel B. Mowrey Faculty Fellow Georgia Institute of Technology
Dec. 2017–Aug. 2022	**University of California, Berkeley**, Berkeley, CA Postdoctoral Researcher, Department of Chemistry / Kavli Energy NanoScience Institute Advisor: A. Paul Alivisatos; Collaborator: Kranthi K. Mandadapu

Interdisciplinary Appointments and Affiliations

- Affiliated Faculty, Institute for Matter and Systems, Georgia Tech.
- Affiliated Faculty, Institute for Data Engineering and Science, Georgia Tech.
- Affiliated Faculty, Institute for Bioengineering and Biosciences, Georgia Tech.
- Program Faculty, Machine Learning Ph.D. Program, Georgia Tech.
- Program Faculty, Bioengineering Ph.D. Program, Georgia Tech.
- Member, Strategic Energy Institute, Georgia Tech.

EDUCATION

2017	**Ph.D., Chemical and Biomolecular Engineering** — Rice University, Houston, TX Advisor: Matteo Pasquali Thesis: *Morphology of Carbon Nanotube Liquid Crystalline Solutions: Insights into Tactoids and Columnar Phase*; Committee: Paul van der Schoot, S. Lisa Biswal, Fred C. MacKintosh
2011	**B.Sc., Chemical Engineering** — Sharif University of Technology, Tehran, Iran

HONORS AND AWARDS

Research Honors, Fellowships, and Professional Recognition

2026	Digital Discovery Emerging Investigator
2025	Beckman Young Investigator Award finalist
2024–2026	Scialog Fellow, Automating Chemical Laboratories, Research Corporation for Science Advancement
2024	National Science Foundation CAREER Award
2023	ACS Petroleum Research Fund Doctoral New Investigator Award

2023	AI for Chemicals and Materials Discovery Initiative Lead, Georgia Tech
2023	35 Voices of Chemistry of Materials, American Chemical Society
2021	Rising Stars in Soft and Biological Matter, University of Chicago MRSEC
2020	AIChE Women in Chemical Engineering Travel Award
2019	Future Faculty in Soft Matter Workshop, Princeton University and University of Delaware
2019	ACS P2F Future Faculty Scholar
2017	Society of Rheology Student Travel Award
2016	Active and Smart Matter Conference Travel Award
2016	Society of Iranian-American Women for Education Scholarship
2014	ConocoPhillips Endowed Scholarship
2014	Phi Lambda Upsilon Honor Society
2013	NASA Space Health Innovation Challenge hackathon finalist

Institutional, Teaching, and Student Recognition

2024	Georgia Tech Thank-a-Teacher recognition
2021	Berkeley Postdoctoral Association Professional Development Award
2016	Smalley-Curl Institute Travel Award, Rice University
2013	Ignite Silicon Valley Trek Travel Award, Rice Alliance for Technology and Entrepreneurship
2012	Best Teaching Assistant Award, Department of Chemical and Biomolecular Engineering, Rice University
2012	Screech Elevator Pitch Competition People's Choice Award, Rice Center for Engineering Leadership

PUBLICATIONS

* denotes equal contribution; † denotes corresponding author.

1. Panicker, I. A., Requejo, K. I., Ruiz, R., Ben-Moshe, A., Bustamante, C., Alivisatos A. P., **Jamali, V.**†, "Polarity-resolved imaging reveals active transport in catalytic Janus nanomotors", *ChemRxiv*, 2026. (under review at *PNAS*)
2. Patel, R., Briceno, G., Aghazadeh, A., **Jamali, V.**†, "Multimodal LLM-assisted diagnosis of cryptic failure modes in high-throughput experimentation: A case study in gold nanorod synthesis", *Neural Information Processing Systems, AI for Accelerated Materials Design Workshop*, 2026.
3. Shabeeb, Z.*, Sreeram, S.*, Panicker, I. A., Wang, A., Xu, M., Goel, R., **Jamali, V.**†, "SAM-EM: Segment Anything for Liquid Phase Electron Microscopy Video Segmentation", *Digital Discovery*, 5(9), 3625–3635 (2026), Emerging Investigators Special Collection. Website: https://jamalilab.github.io/SAM-EM/
4. Shabeeb, Z.*, Saeedi, D.*, Tsui, D.*, **Jamali, V.**†, Aghazadeh, A.†," Generative Priors for Cryo-EM Image Reconstruction", *ICML Workshop on Generative and Agentic AI for Biology* (2026).
5. **Jamali, V.,**†, Aghazadeh, A.†, Kacher, J., "Thinking Microscopes: Agentic AI and the Future of Electron Microscopy", *npj Computational Materials* 12, 149, (2026).
6. Panicker I.A., Shabeeb, Z., Hargus, C., **Jamali, V.**†,"Solution-Tunable Interfacial Interaction Landscape Governs Anomalous Nanoparticle Diffusion in Liquid-Phase Electron Microscopy". *ACS Nano* 20, 21, 15550–15564

(2026) (Featured on front cover).

7. Requejo, K. I., Cañari-Chumpitaz, C., **Jamali, V.**, Ruiz, R., Alivisatos, A. P., Bustamante, C. "Quantifying photochemical propulsion in light-powered Janus micromotors". *ACS Nano* 20, 19, 14099–14112 (2026).
8. Shabeeb, Z.*, Saeedi, D.*, Tsui, D.*, **Jamali, V.**†, Aghazadeh, A.†," cryoSENSE: Compressive Sensing Enables High-throughput Microscopy with Sparse and Generative Priors on the Protein Cryo-EM Image Manifold". *IEEE/CVF, Computer Vision and Pattern Recognition (CVPR)*, (2026). Website: https://cryosense.github.io
9. Housley S., N.†, Bourque, A.R., Matyunina, L.V., Panicker, I.A., Schrader Echeverri, E., Izykowicz, M., Herrmann, O.A., Lee, S., Arboleda, J.C., **Jamali, V.**, McDonald, J.F., Dahlman, J. E., Finn, M.G.,"Tumor Agnostic Drug Delivery with Dynamic Nanohydrogels". *Nature Communications*, 17, 184 (2026).
10. Shabeeb, Z., Goyal, N., Attah Nantogmah, P., **Jamali, V.**†,"Learning the Diffusion of Nanoparticles in Liquid Phase TEM via a Physics-informed Generative AI". *Nature Communications* 16, 6298 (2025).
11. Shabeeb, Z., **Jamali, V.**†,"LEONARDO: A Physics-Informed Generative Model for Stochastic Nanoparticle Dynamics in Liquid-Phase TEM". *Neural Information Processing Systems, AI for Science Workshop* (2025).
12. Wang*, A., Xu*, M., Goel, R., Shabeeb, Z., Panicker, I. A., **Jamali, V.**†, "SAM-EM: Real-Time Segmentation for Automated Liquid Phase Transmission Electron Microscopy", *Neural Information Processing Systems, AI for Accelerated Materials Design Workshop* (2025).
13. Shabeeb, Z., Goyal, N., Attah Nantogmah, P. **Jamali, V.**†,"Learning the Physics of Liquid Phase TEM Nanoparticle Trajectories Using Physics-Informed Generative AI". *Microscopy and Microanalysis* 30, S1, 1724 (2024).
14. Shabeeb, Z., Goyal, N., **Jamali, V.**†, "Determining Diffusion Characteristics of Nanoparticles in Liquid Phase TEM Using Deep Learning". *Microscopy and Microanalysis* 30, S1, 2074 (2024).
15. **Jamali, V.**†, Alivisatos, A. P.†, "Studying Diffusion of Colloidal Nanoparticles in Solution Using Liquid Phase TEM and Machine Learning". *Microscopy and Microanalysis* 28, 142 - 143 (2022).
16. Moreno-Hernandez, I. A.*, Crook, M. F.,* **Jamali, V.***, Alivisatos, A. P., "Recent advances in the study of colloidal nanocrystals enabled by in situ liquid phase transmission electron microscopy". *MRS Bulletin* 47 (3), (2022).
17. Abbas, A., Vargo, E., **Jamali, V.**, Ercius, P., Pieters, P., Brinn, R., Ben-Moshe, A., Cho, M., Xu, T., Alivisatos, A. P. "Observation of an Orientational Glass in a Superlattice of Elliptically-Faceted CdSe Nanocrystals". *ACS Nano* (2022) 16(6), 9339–9347.
18. **Jamali, V.**, Hargus, C., Ben Moshe A., Aghazadeh, A., Ha, H. D., Mandadapu, K. K., Alivisatos, A. P. "Deep Learning-assisted Liquid Cell Electron Microscopy Reveals the Nature of Anomalous Diffusion of Nanoparticles Near the Surface". *Proceedings of the National Academy of Sciences (PNAS)* 118 (10) (2021).
19. **Jamali, V.**, Mirri, F., Biggers, E. G., Pinnick, R.A., Liberman, L., Cohen, Y., Talmon, Y., MacKintosh F., van der Schoot, P., Pasquali, M. "Enhanced ordering in length-polydisperse carbon nanotube solutions at high concentrations as revealed by the small angle X-ray scattering". *Soft Matter* 17, 5122-5130 (2021).
Featured on the front cover of Soft Matter, Issue 20.
20. Cho, H., Moreno-Hernandez, I., **Jamali, V.**, Oh, M., Alivisatos, A. P. "In situ quantification of interactions between charged nanorods in a predefined potential energy landscape". *Nano Letters* 21 (1), 628-633 (2021).
21. **Jamali, V.***, Niroui, F.*, Taylor, L. W., Dewey, O. S., Koscher, B. A., Pasquali, M., Alivisatos, A. P. "Perovskite-Carbon Nanotube Light Emitting Fibers". *Nano Letters* 20 (5), 3178-3184 (2020).
22. Liberman, L., **Jamali, V.**, Pasquali, M., Talmon, Y. "The effect of carbon nanotube diameter and stiffness on their phase behavior in crowded solutions". *Langmuir* 36 (1), 242-249 (2020).
23. Mirri, F.*, Ashkar, R.*, **Jamali, V.**, Liberman, L., Pinnick, R., Talmon, Y., van der Schoot, P., Butler, P., Pasquali, M. "Quantification of carbon nanotube liquid crystal morphology via neutron scattering". *Macromolecules* 51

(17), 6892-6900 (2018).

24. Maillaud, L., Headrick, R. J., **Jamali, V.**, Maillaud, J., Tsentalovich, D., Neri, W., Bengio, E. A., Mirri, F., Kleinerman, O., Talmon, Y., Poulin, P., and Pasquali, M., "Flexible and conductive fibers made from highly concentrated aqueous dispersions of carbon nanotubes". *Industrial and Engineering Chemistry Research* 57 (10), 3554-3560 (2018).
25. Tran, T. Q., Headrick, R. J., Bengio, E. A., Myint, S. M., Khoshnevis, H., **Jamali, V.**, Duong, H. M., Pasquali, M. "Purification and dissolution of carbon nanotube fibers spun from floating catalyst method". *ACS Applied Materials and Interfaces* 9 (42), 37112-37119 (2017).
26. **Jamali, V.**, Biggers, E., van der Schoot, P., Pasquali, M. "Line tension of twist-free carbon nanotube lyotropic liquid crystal microdroplets on solid surfaces". *Langmuir* 33 (36), 9115-9121 (2017).
27. Jiang, C., Peng, Z., de los Reyes, C., Young, C. C., Tsentalovich, D., **Jamali, V.**, Ajayan, P. M., Tour, J. M., Pasquali, M., and Marti A. A., "Increased solubility and fiber spinning of graphenide dispersions aided by crown-ethers". *Chemical Communications* 53 (9), 1498-1501 (2016).
28. **Jamali, V.***, Behabtu, N.*, Senyuk, B., Lee, J. A. Smalyukh, I.I., van der Schoot, P., Pasquali, M. "Experimental realization of crossover in shape and director field of nematic tactoids". *Physical Review E* 91 (4), 042507 (2015).

PATENTS AND INTELLECTUAL PROPERTY

1. Aghazadeh, A., Jamali, V., Saeedi, D., Shabeeb, Z., Tsui, D., "Compressive Sensing Enables High-throughput Microscopy with Sparse and Generative Priors on the Protein Cryo-EM Image Manifold", U.S. provisional application USSN 63/977,630.
2. Alivisatos, A. P., Niroui, F., Jamali, V., Pasquali, M., "Light Emitting Fibers", U.S. provisional application USSN 62/714,561.
3. Alivisatos, A. P., Jamali, V., "Processing Method for Fabricating Perovskite-Carbon Nanotube Fibers and Devices", U.S. provisional application USSN 62/958,394.

RESEARCH SOFTWARE AND OPEN-SOURCE RESOURCES

1. **cryoSENSE**: Compressive sensing with sparse and generative priors on the protein cryo-EM image manifold. Project website.
2. **SAM-EM**: Segment Anything for liquid-phase electron microscopy video segmentation. project website; Project website.
3. **LEONARDO**: Physics-informed generative modeling of nanoparticle diffusion in liquid-phase TEM. Project website.
4. **MotionNet / MoNet**: Physics-informed deep learning for classifying single-particle trajectories and estimating the anomalous diffusion exponent. Project website.

FUNDED RESEARCH GRANTS AND SUPPORT

Principal Investigator

2026–2027 **Research Corporation for Science Advancement**—*Closing the Characterization Loop in Autonomous Materials Discovery: Agentic AI for Multimodal Instrument Orchestration*.
PI; direct funding $66,000; Aug. 1, 2026–July 30, 2027.

2026–2027 **Foresight Institute**—*Self-Driving Nanofabrication with Thinking Microscopes*.
PI; direct funding $10,000; Oct. 1, 2026–Sept. 30, 2027.

2026–2027 **Microsoft Azure**—*Experiment-in-the-Loop Agentic AI for Task-Adaptive Self-Driving Labs*.
PI; $12,500 in computing support; Oct. 1, 2026–Sept. 30, 2027.

2024–2028 **National Science Foundation (NSF CAREER)**—*CAREER: Imaging and Understanding the Motion and Interaction of Nanoparticles Near Surfaces*.
PI; total award $744,338; Jan. 1, 2024–Dec. 30, 2028.

2024–2026 **Georgia Tech Institute for Electronics and Nanotechnology**—*Compressed Super-Resolution TEM Using a Nanoelectronic Coded Aperture Device*.
PI; co-PI: Amirali Aghazadeh; total award $225,000 in direct costs; Jan. 1, 2024–Dec. 30, 2026.

2023–2025 **American Chemical Society Petroleum Research Fund (DNI)**—*Investigating Rotational and Translational Diffusion of Nanoparticles at Liquid–Liquid Interfaces via Time-Resolved Nanoscale Imaging*.
PI; total award $110,000; Sept. 1, 2023–Aug. 31, 2025.

Co-Principal Investigator

2024 **Georgia Tech Institute for Data Engineering and Science and Institute for Materials**—*AI for Chemicals and Materials Discovery*.
Co-PI; collaborators: Victor Fung, Pan Li, Amirali Aghazadeh; total award $20,000 Jan. 1–Dec. 1, 2024.

INVITED TALKS AND RESEARCH PRESENTATIONS

Upcoming Invited Talks

Apr 2027 **LPTEM Reveals Active Transport and Polarity-Dependent Propulsion of Catalytic Janus Nanomotors** *(invited)*
MRS Spring Meeting, Seattle, WA

Invited Conference, Symposium, and Workshop Talks

Sep 2026 **AI-Driven Electron Microscopy: Imaging, Learning, and Understanding Nanoscale Dynamics** *(invited)*
Venice meeting on fluctuations in small complex systems VIII, Venice, Italy

Jan 2026 **AI-Assisted LPTEM: A Unified Framework for Nanoparticle Tracking and Generative Modeling** *(invited)*
Gordon Research Conference on Liquid Phase EM, Ventura, CA

May 2025 **On the Viscoelastic Origin of Nanoparticle Surface Diffusion in LPTEM** *(invited)*
Southeast Soft Matter Day, Atlanta, GA

Apr 2025 **Imaging, Learning, and Engineering of Nanoscale Dynamics via LPTEM** *(invited)*
MRS Spring Meeting, Seattle, WA

Sep 2023 **Using Deep Learning to Elucidate the Dynamics of Nanoparticles in LPTEM** *(invited)*
International Microscopy Congress IMC20, Busan, Korea

Mar 2023 **Deep Learning-Assisted Analysis of Anomalous Nanoparticle Surface Diffusion in LPTEM** *(invited)*
2023 Institute for Materials Symposium on Materials Innovations, Atlanta, GA

Mar 2023 **Deep Learning-Assisted Analysis of Anomalous Nanoparticle Surface Diffusion in LPTEM** *(invited)*
CCMST, Machine Learning in Chemistry Seminar, GeorgiaTech, Atlanta, GA

Oct 2022 **Deep Learning-Assisted Analysis of Anomalous Nanoparticle Surface Diffusion in LPTEM** *(invited)*
LPTEM Gordon Research Conference, Ventura, CA

Aug 2022 **Studying Diffusion of Colloidal Nanoparticles in Solution Using LPTEM and Machine Learning** *(invited)*
Microscopy and Microanalysis Conference, Portland, OR

Sep 2021 **Imaging, learning, and engineering of complex soft matter systems at the nanoscale**
Rising Stars in Soft and Biological Matter Symposium, University of Chicago MRSEC (virtual)

Mar 2018 **Morphology of carbon nanotube liquid crystalline phases: insight into tactoids and columnar phase** *(invited)*
APS March Meeting, Los Angeles, CA

Invited Departmental Seminars, Institutional Lectures, and Webinars

July 2026 **AI-Driven Electron Microscopy: Imaging, Learning, and Understanding Nanoscale Dynamics** *(invited)*
Electron Microscopy sub-group, Materials Science and Engineering, Stanford University

May 2026 **AI-Driven Electron Microscopy for Visualizing Nanoscale Dynamics in Liquid** *(invited)*
NSF Center for Chemical Innovation (CCI), University of Chicago, Chicago, IL (virtual)

Jan 2026 **Decoding Nanoscale Motion: AI-Driven Liquid Phase TEM** *(invited)*
Protochips Webinar on Exploring Synthesis of Nanomaterials Using Liquid Phase TEM (Virtual)

Dec 2025 **AI-Driven Electron Microscopy for Visualizing Nanoscale Dynamics in Liquid** *(invited)*
University of California, Santa Barbara, CA

Oct 2025 **AI for Anomalous Diffusion: From Classification to Generative Modeling of Stochastic Transport at the Nanoscale**
Potsdam University, Potsdam, Germany (Invited, Virtual)

May 2025 **Transforming electron microscopes to automated factories of discovery**
Materials x Robotics Initiative, Institute for Matter and Systems, Georgia Tech, Atlanta, GA

Nov 2024 **Learning the Diffusion of Nanoparticles in LPTEM Using Physics-Informed Generative AI** *(invited)*
Ohio State University-Protochips Webinar on Liquid Cell TEM, Columbus, OH

Mar 2024 **Imaging, Learning, and Engineering of Nanoparticles' Dynamics Using LPTEM** *(invited)*
Institute for Electronics and Nanotechnology, Georgia Tech, Atlanta, GA

Mar 2022 **Imaging, learning, and engineering of soft matter systems at the nanoscale** *(invited)*
University of Southern California, Chemical Engineering and Materials Science, Los Angeles, CA

Feb 2022 **Imaging, learning, and engineering of soft matter systems at the nanoscale** *(invited)*
Massachusetts Institute of Technology, Department of Chemical Engineering, Cambridge, MA

Feb 2022 **Imaging, learning, and engineering of soft matter systems at the nanoscale** *(invited)*
Brandeis University, MRSEC, Waltham, MA

Feb 2022 **Imaging, learning, and engineering of soft matter systems at the nanoscale** *(invited)*
University of Minnesota, Department of Chemical Engineering and Materials Science, Minneapolis, MN

Feb 2022 **Imaging, learning, and engineering of soft matter systems at the nanoscale** *(invited)*
University of Wisconsin-Madison, Department of Chemistry, Madison, WI

Feb 2022 **Imaging, learning, and engineering of soft matter systems at the nanoscale** *(invited)*
University of California, Los Angeles, Department of Chemistry and Biochemistry, Los Angeles, CA

Feb 2022 **Imaging, learning, and engineering of soft matter systems at the nanoscale** *(invited)*
Yale University, Department of Chemical and Environmental Engineering, New Haven, CT

Jan 2022 **Imaging, learning, and engineering of soft matter systems at the nanoscale** *(invited)*
Cornell University, Department of Chemistry and Chemical Biology, Ithaca, NY

Jan 2022 **Imaging, learning, and engineering of soft matter systems at the nanoscale** *(invited)*
Princeton University, Department of Chemical and Biological Engineering, Princeton, NJ

Jan 2022 **Imaging, learning, and engineering of soft matter systems at the nanoscale** *(invited)*
Georgia Institute of Technology, School of Chemical and Biomolecular Engineering, Atlanta, GA

Dec 2021 **Imaging, learning, and engineering of soft matter systems at the nanoscale** *(invited)*
University of California, Berkeley, Kavli Energy NanoScience Institute, Berkeley, CA

Sep 2021 **Deep learning-assisted analysis of anomalous nanoparticle surface diffusion in liquid phase TEM** *(invited)*
University of California, Berkeley, Nano Seminar Series, Berkeley, CA

Jul 2021 **Imaging, learning, and engineering of complex soft matter systems at the nanoscale** *(invited)*
Seagate Normandale AI/ML Distinguished Seminar Series, Bloomington, MN (virtual)

Feb 2021 **Imaging, learning, and engineering of complex soft matter systems at the nanoscale**
Department of Chemical and Biomolecular Engineering, Johns Hopkins University, MD (virtual)

Feb 2021 **Imaging, learning, and engineering of complex soft matter systems at the nanoscale**
Department of Chemical Engineering, Texas A&M University, TX (virtual)

Jan 2021 **Imaging, learning, and engineering of complex soft matter systems at the nanoscale**
Department of Chemical and Biomolecular Engineering, University of Notre Dame, IN (virtual)

Feb 2019 **From carbon nanotube liquid crystalline solutions to functional fibers** *(invited)*
Department of Materials Science and Engineering, Cornell University, Ithaca, NY

Nov 2018 **From carbon nanotube liquid crystalline solutions to functional fibers** *(invited)*
Department of Physics, Physics of Living Systems, MIT, Cambridge, MA

Feb 2017 **Phase behavior and morphology of carbon nanotube liquid crystal solutions** *(invited)*
Lewis-Sigler Integrative Genome Institute, Biophysics group, Princeton, NJ

Other Conference and Workshop Presentations

Dec 2025 **SAM-EM: Real-Time Segmentation for Automated Liquid Phase Transmission Electron Microscopy [Spotlight]**
AI4Mat workshop, Neural Information Processing Systems, San Diego, CA

Aug 2025	**Liquid phase transmission electron microscopy for investigating multiscale dynamics of biomolecular systems** 1st SouthEast Amyloid-CONdensate (SEACON) symposium, Ocean Springs, MS
Jun 2025	**AI-enabled Single Particle Tracking in Liquid-Phase Transmission Electron Microscopy** ACS Colloid & Surface Science Symposium, Alberta, Canada
Mar 2025	**Physics-informed Generative Models for Learning the Stochastic Diffusion of Single Particle Trajectories** American Physical Society March Meeting, Anaheim, CA
Oct 2024	**Liquid Phase Transmission Electron Microscopy for Single Particle Tracking** AIChE Fall Meeting, San Diego, CA
Mar 2024	**Deep Generative Neural Networks for Learning the Dynamics of Nanoparticles in LPTEM** APS March Meeting, Minneapolis, MN
Nov 2021	**Deep Learning-Assisted Analysis of Anomalous Nanoparticle Surface Diffusion in Liquid Phase TEM** AIChE Annual Meeting, Boston, MA
Nov 2021	**Deep learning-assisted analysis of anomalous nanoparticle diffusion near the liquid cell surface reveals the effect of electron beam dose rate in TEM** AIChE Annual Meeting, Boston, MA
Jun 2021	**Deep learning-assisted analysis of anomalous nanoparticle surface diffusion in liquid phase TEM** ACS Colloid & Surface Science Symposium (virtual)
Mar 2021	**Deep learning-assisted analysis of anomalous nanoparticle diffusion near the liquid cell surface reveals the effect of electron beam dose rate in TEM** APS March Meeting (virtual)
Nov 2020	**In-situ liquid phase electron microscopy for studying the dynamics of colloidal nanoparticles at the nanoscale** AIChE Annual Meeting, San Francisco, CA (virtual, available online)
Nov 2019	**From nanoscale building blocks to functional fibers** AIChE Annual Meeting, Orlando, FL
Nov 2018	**Colloidally synthesized nanomaterials as building blocks for functional fibers** MRS Fall Meeting, Boston, MA
Oct 2018	**From carbon nanotube liquid crystalline solutions to functional fibers** AIChE Annual Meeting, Pittsburgh, PA
Oct 2017	**A hexagonal columnar liquid crystal phase formation in dilute solutions of carbon nanotubes** AIChE Annual Meeting, Minneapolis, MN
Feb 2017	**Phase behavior and morphology of carbon nanotube liquid crystal solutions** 88th Society of Rheology Conference, Tampa, FL
Nov 2016	**Morphology of carbon nanotube liquid crystal solutions** AIChE Annual Meeting, San Francisco, CA

Jun 2016 **Wetting behavior, shape, and morphology of sessile lyotropic liquid crystal microdroplets**
ACS Colloid & Surface Science Symposium, Harvard University, Cambridge, MA

Apr 2016 **Wetting behavior, shape, and morphology of sessile lyotropic liquid crystal microdroplets**
SCI Transdisciplinary Symposium, Rice University, Houston, TX

Aug 2015 **Experimental realization of crossover in shape and director field of nematic tactoids**
Texas Soft Matter Meeting, Houston, TX

Jun 2015 **Experimental realization of crossover in shape and director field of nematic tactoids**
ACS Colloid & Surface Science Symposium, Carnegie Mellon University, Pittsburgh, PA

TEACHING AND EDUCATIONAL CONTRIBUTIONS

Georgia Tech Courses — Instructor of Record

Fall 2026 **AI for Experimental Chemical Engineers** (CHBE 4803/8803); 28 students.

Spring 2026 **Fluid Mechanics** (CHBE 3205); 53 students.

Fall 2025 **Fluid Mechanics** (CHBE 3205); 40 students.

Spring 2025 **AI for Experimental Chemical Engineers** (CHBE 4803/8803); 19 students.

Fall 2024 **Fluid Mechanics** (CHBE 3205); 21 students.

Spring 2024 **Fluid Mechanics** (CHBE 3205); 44 students.

Fall 2023 **Transport Phenomena I** (CHBE 3200); 44 students.

Fall 2022 **Transport Phenomena I** (CHBE 3200); 33 students.

Earlier Teaching Experience — Rice University

Fall 2014 Dean's Teaching Assistant, Thermodynamics I (CHBE 411).

Spring 2014 Teaching Assistant, Colloidal and Interfacial Phenomena (CHBE 560).

Fall 2012 Teaching Assistant, Transport Phenomena I (CHBE 401).

Fall 2011 Teaching Assistant, Chemical Engineering Laboratory II (CHBE 433).

Course Development and Teaching Leadership

- Designed and introduced *AI for Experimental Chemical Engineers* in Spring 2025, a course associated with the Applications of AI and ML in Chemical Engineering minor and the Graduate Data Science Certificate.
- Awarded a College of Engineering AI Makerspace grant for course development.
- Undergraduate faculty advisor to sophomores, School of Chemical and Biomolecular Engineering, Georgia Tech (2023–present).

RESEARCH MENTORING AND STUDENT GUIDANCE

Ph.D. Research Advisees

Fall 2026–present **Ziyi (Lucy) Liu**, ChBE, Georgia Tech.
Co-advised with Amirali Aghazadeh and Lily Cheung. Tentative thesis: *Liquid–Liquid Phase-Separated Peptide Coacervates for Sustainable Coatings*.

Fall 2026–present	**Tianhua Zhao**, ChBE, Georgia Tech. Tentative thesis: *Multi-Agent AI for Planning and Control of Self-Driving Labs*.
Fall 2025–present	**Yuling Zheng**, Bioengineering, Georgia Tech. Passed qualifying examination. Tentative thesis: *Cryogenic and Liquid-Phase Electron Microscopy Methods for Imaging Protein Macromolecules and Their Assemblies*.
Fall 2023–present	**Isabel Panicker**, ChBE, Georgia Tech. Passed Ph.D. proposal. NSF GRFP honorable mention; Eckert Fellowship. Tentative thesis: *A Correlative Microscopy Platform for Studying Non-Equilibrium Dynamics of Active Matter*.
Fall 2022–present	**Zain Shabeeb**, ChBE, Georgia Tech. Passed Ph.D. thesis proposal. GT LEAD Fellow; Eckert Fellowship. Tentative thesis: *AI-driven Electron Microscopy for Probing Nanoscale Biological Dynamics*.

Undergraduate Research Trainees — Georgia Tech

Fall 2026–present	**Milan Barnes**, ChBE.
Fall 2026–present	**Sardhak Boddu**, ChBE.
Fall 2026–present	**Neha Muralitharan**, ChBE.
Fall 2026–present	**Caroline Tzur**, ChBE.
Fall 2026–present	**Benjamin Brough**, ChBE.
Fall 2026–present	**Nathaniel (Nate) Glanzmann**, ChBE.
Summer–Fall 2026	**Gabriela Briceno**, Materials Science and Engineering.
Spring 2026–present	**Riya Patel**, ChBE.
Fall 2025–present	**Siddarth Sreeram**, Computer Engineering.
Spring 2025–present	**Diya Mahesh**, ChBE.
Summer 2026	**Mahika Raj**, Computer Science.
Spring–Summer 2026	**Anika Patel**, Computer Science.
Summer 2025–Summer 2026	**Alexander Wang**, Computer Science.
Summer 2025–Spring 2026	**Max Xu**, Computer Science.
Summer–Fall 2025	**Madison Tat**, ChBE; Presidential Undergraduate Research Award (PURA) fellow.
Fall 2024–Spring 2025	**Risha Goel**, ChBE; PURA fellow; now Ph.D. student at the University of Minnesota.
Fall 2024–Spring 2025	**Laurence Lines**, ChBE.
Fall 2024–Spring 2025	**Isabella Faciano**, ChBE.
Fall 2024–Spring 2026	**Lauren Caldwell**, ChBE.
Fall 2024	**Matthew Abernathy**, ChBE.
2023–2025	**Cecilia Reed**, ChBE.
Fall 2023	**Gabriel Joaquim Sampaio de Almeida**, ChBE.
Fall 2023	**Arko Roy**, ChBE.
Fall 2022–Spring 2024	**Naisargi Goyal**, ChBE; PURA fellow; now Ph.D. student at Princeton University.

Earlier Research Trainees and Interns

2018–2019	**Tanner Yamada**, University of California, Berkeley, Chemical and Biomolecular Engineering.
2016–2017	**Evan Biggers**, Rice University, Chemical and Biomolecular Engineering.
Summer 2017	**Miranda Mittleman**, high school summer intern, Rice University.
2016–2017	**Samuel Quitzau**, NSF Research Experiences for Undergraduates (REU), Rice University.

Graduate Thesis and Dissertation Committees

2026–present	**Ahmed Mohamed**, Materials Science and Engineering; advisors: Natalie Stingelin and Juan Pablo Correa-Baena.
2026–present	**Seongeun Kim**, ChBE; advisors: Lily Cheung and Shu Takayama.
2025–present	**Evan Moore**, ChBE; advisor: Victor Breedveld.
2024–present	**Sayan Bhowmik**, ChBE; advisor: AJ Medford.
2024–2026	**Jiaji Lin**, ChBE; advisor: Krista Walton; defended.
2023–2025	**Myeongsoo Kim**, Bioengineering; advisor: Stanislav Emelianov; defended.
2024	**Sushree Jagriti Sahoo**, ChBE; advisor: AJ Medford; defended.
2024–2025	**Sydney Wimberley**, ChBE; advisor: Julie Champion; defended.
2023–2026	**Natechanok Yutthasaksunthorn**, ChBE; advisor: Sankar Nair; defended.
2023–2025	**Zhitao Chen**, ChBE; advisor: Nian Liu; defended.
2023	**Ifayoyinsola A. Ibikunle**, ChBE; advisor: David Sholl; defended.

PROFESSIONAL SERVICE, SCIENTIFIC LEADERSHIP, AND OUTREACH

Conference Organization and Session Leadership

2027	Organizer and chair, DSOFT/DBIO Mini-Symposium, APS Global Physics Summit.
2026	Session organizer and chair, AIChE Fall Meeting, Nanoscience and Engineering Forum.
2025	Session organizer and chair, AIChE Fall Meeting, Nanoscience and Engineering Forum.
2025	Session chair, ACS Colloid and Surface Science Symposium.
2024	Session organizer and chair, AIChE Fall Meeting, Nanoscience and Engineering Forum.
2024	Organizer, Atlanta Soft Matter Day Symposium.
2023	Session co-chair, International Microscopy Congress.
2022	Session co-chair, Microscopy and Microanalysis.
2017	Session co-chair, AIChE Annual Meeting.

Editorial, Refereeing, and Grant Review Service

2022–present	Guest Editor, *Journal of Visualized Experiments* (JoVE).
	Journal reviewer: *Science*; *Science Advances*; *Journal of the American Chemical Society*; *ACS Nano*; *Soft Matter*; *Physical Review E*; *PRX Life*; *ACS Applied Electronic Materials*; *ACS Applied Polymer Materials*; *Journal of Rheology*; *Physical Chemistry Chemical Physics*; *Nanoscale Horizons*; *Digital Discovery*; *Advanced Intelligent Systems*.
	Conference program committee/reviewer: NeurIPS AI4Mat; ICML AI4Mat.

2024	Panelist, NSF CAREER, CBET program.
2021	Reviewer, Graduate Women in Science Fellowship.

Georgia Tech Committee Service

2026–present	Information Technology Committee, School of ChBE.
2025–present	Faculty Search Committee, School of ChBE.
2024–2026	ChBE Advisory Committee, School of ChBE.
2024–2025	Graduate Admissions Committee, School of ChBE.
2024–present	Facilities Advisory Committee, Institute for Matter and Systems.
2022–2025	Seminar Organization Committee, School of ChBE.
2022–2024	Graduate Recruitment Committee, School of ChBE.
2023–present	Sophomore undergraduate faculty advisor, School of ChBE.
2023	Bioengineering Day Thesis Competition Judge.

Research and Academic Program Development

2023–2025	Co-lead, AI for Chemicals and Materials Discovery Initiative, Institute for Data Engineering and Science, Georgia Tech.
2024–present	Research program lead, Compressed Super-Resolution TEM Using a Nanoelectronic Coded Aperture Device, Georgia Tech.
	Program faculty: Machine Learning Ph.D. Program and Bioengineering Ph.D. Program, Georgia Tech.
	Institute affiliations: Institute for Matter and Systems; Institute for Bioengineering and Biosciences; Institute for Data Engineering and Science; Strategic Energy Institute.

Professional Society Memberships

- American Institute of Chemical Engineers (AIChE).
- American Chemical Society (ACS).
- American Physical Society (APS).
- Materials Research Society (MRS).
- Microscopy Society of America (MSA).

Outreach and Public Engagement

2026	Hosted K–12 teacher through NNCI Research Experience for Teachers (RET) program, Jamali Group.
2025–2026	DyeNamic Lab, Atlanta Science Festival, Atlanta, GA.
	Hosted tours of the Georgia Tech electron microscopy facility for Campbell and Walton High School students.
2024	Panelist, UC Berkeley Visiting Scholar and Postdoctoral Association, Academic Job Search Strategies Workshop.

2023	Panelist, NextProf program.
2023	Panelist, Georgia Tech AIChE Women in Chemical Engineering.
2023	STEM GEM workshop panelist/presenter, Atlanta, GA.
2022	Postdoctoral mentorship and career guidance sessions, Gordon Research Conference on Liquid-Phase Electron Microscopy.
2019–2020	Postdoctoral representative, Chemistry Graduate Life Committee, UC Berkeley.
2013–2014	Recitation chair, Chemical and Biomolecular Engineering Graduate Student Association, Rice University.
2012, 2013	Workshop presenter, Sally Ride Science Festival for Girls, Houston, TX.

SELECTED RESEARCH COVERAGE AND PRESS

Jun. 5, 2026	**Georgia Tech**: Study Uncovers New Insights into Nanoparticle Motion
May 22, 2026	**Phys.org**: Agentic AI Could Help Electron Microscopes Plan, Adapt, and Analyze Experiments
May 18, 2026	**Georgia Tech**: Accelerating Discovery with AI
Mar. 10, 2026	**Medical Xpress**: Nanohydrogels Steer Cancer Drugs to Tumors, Aiming to Spare Healthy Tissue
Mar. 6, 2026	**Georgia Tech**: Target the Tumor. Spare the Body
July 15, 2025	**Phys.org**: New AI Tool Deciphers Mysteries of Nanoparticle Motion in Liquid Environments
July 11, 2025	**Georgia Tech**: New AI Tool Deciphers Mysteries of Nanoparticle Motion in Liquid Environments
May 31, 2021	**Rice University**: Molecular Jiggling Has Implications for Carbon Nanotube Fibers
May 31, 2021	**Phys.org**: Molecular Jiggling Has Implications for Carbon Nanotube Fibers
Nov. 10, 2017	**Materials Today**: Scientists Explore Ways to Produce High-Quality Fibers from Carbon Nanotubes
Oct. 16, 2017	**Phys.org**: Researchers Advance Characterization, Purification of Nanotube Wires and Films
Oct. 16, 2017	**EurekAlert!**: Long Nanotubes Make Strong Fibers

PROFESSIONAL REFERENCES

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